

HelloWorld.S

```
@
@ Assembler program print out "Hello World"
@ using the Pico SDK.
@
@ R0 - first parameter to printf
@ R1 - second parameter to printer
@ R7 - index counter
@

.thumb_func      @ Necessary because sdk uses BLX
.global main     @ Provide program starting address to linker

main:
    MOV R7, #0    @ initialize counter to 0
    BL  stdio_init_all @ initialize uart or usb
loop:
    LDR R0, =helloworld @ load address of helloworld string
    ADD R7, #1          @ Increment counter
    MOV R1, R7          @ Move the counter to second parameter
    BL  printf          @ Call pico_printf
    B   loop            @ loop forever

.data
    .align 4           @ necessary alignment
helloworld: .asciz     "Hello World %d U13157004\n"
```